

SalesVerse 2.0

Data Dictionary

Physical data model of the PostgreSQL database: tables, columns, keys, relationships, enumerations and database objects

IIFT Agent/Banca Portal & Back-office for Insurans Islam Family Takaful Sendirian Berhad



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Commercial-in-Confidence. Prepared by iorta TechNXT for Insurans Islam Family Takaful Sendirian Berhad as part of the response to IIFT's Request for Proposal for an Agent/Banca Portal and Back-office solution. Not to be disclosed to third parties without prior written consent.

Document Control

This data dictionary describes every table and column of the SalesVerse 2.0 database as created by the versioned migrations in the source code. It is generated from the Prisma schema and the migration SQL, so it always matches the deployed database. A spreadsheet version with the same content is provided for filtering and searching.

Version history

Version	Date	Author	Description
0.9	September 2026	iorta TechNXT	Internal technical review
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Review and approval

Role	Name	Signature / date
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Document pack

Reference	Document	RFP deliverable	Status
IIFT-SV2-SAD	SalesVerse 2.0 Solution Architecture	DEL-06	Issued
IIFT-SV2-DD	SalesVerse 2.0 Data Dictionary	DEL-07	This document
IIFT-SV2-PSH	SalesVerse 2.0 Production Support Handover	DEL-26, DEL-27, MNT-28	Issued
IIFT-SV2-CSQ	SalesVerse 2.0 Code Standards & Quality Report	DEL-11, NFR-14	Issued
IIFT-SV2-SAR	SalesVerse 2.0 Security Assessment Report	DEL-17, NFR-13	Issued
IIFT-SV2-TST	SalesVerse 2.0 Test Strategy	DEL-12	Issued
IIFT-SV2-TCS	SalesVerse 2.0 Test Cases	DEL-13, DEL-14 (precursor)	Issued

Related material

- SalesVerse-2.0-Data-Dictionary.xlsx: the same content as a workbook, one sheet per table.
- SalesVerse-2.0-Solution-Architecture: data architecture, encryption and retention design.
- Source: apps/api/prisma/schema.prisma and apps/api/prisma/migrations/*/migration.sql.

Conventions

- Monetary amounts are in Brunei dollars (B\$). Times are Brunei time (UTC+8) unless marked UTC.
- RFP requirement identifiers (AP, BO, INT, COM, NFR, DEL, MNT) refer to IIFT's Request for Proposal. COM-01 to COM-10 in this pack are the shared platform requirements of RFP section 4.4.
- Items marked **delivered during implementation** are designed here but not yet present in the working application; everything else describes the application as it runs today.
- Option A, B and C are the commercial options of the proposal: on-premise with a perpetual licence, subscription hosted and managed by iorta on cloud, and source code handover. Prices and commercial

terms are in the proposal and its pricing workbook; this pack quotes a figure only where a technical statement depends on it, and reads it from the same source as the proposal.

- Text in **[square brackets]** is to be completed with IIFT during mobilisation.

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1. Introduction

1.1 Scope

This dictionary covers the complete SalesVerse 2.0 database as defined by migration 20261002133922_initial_schema: 37 tables, 434 columns, 36 enumerated types, 27 foreign keys, 109 indexes including primary keys, 9 sequences, 4 check constraints and one trigger. Schema changes are made only through new migrations, and this document is regenerated with each release that contains one.

1.2 How the dictionary is produced

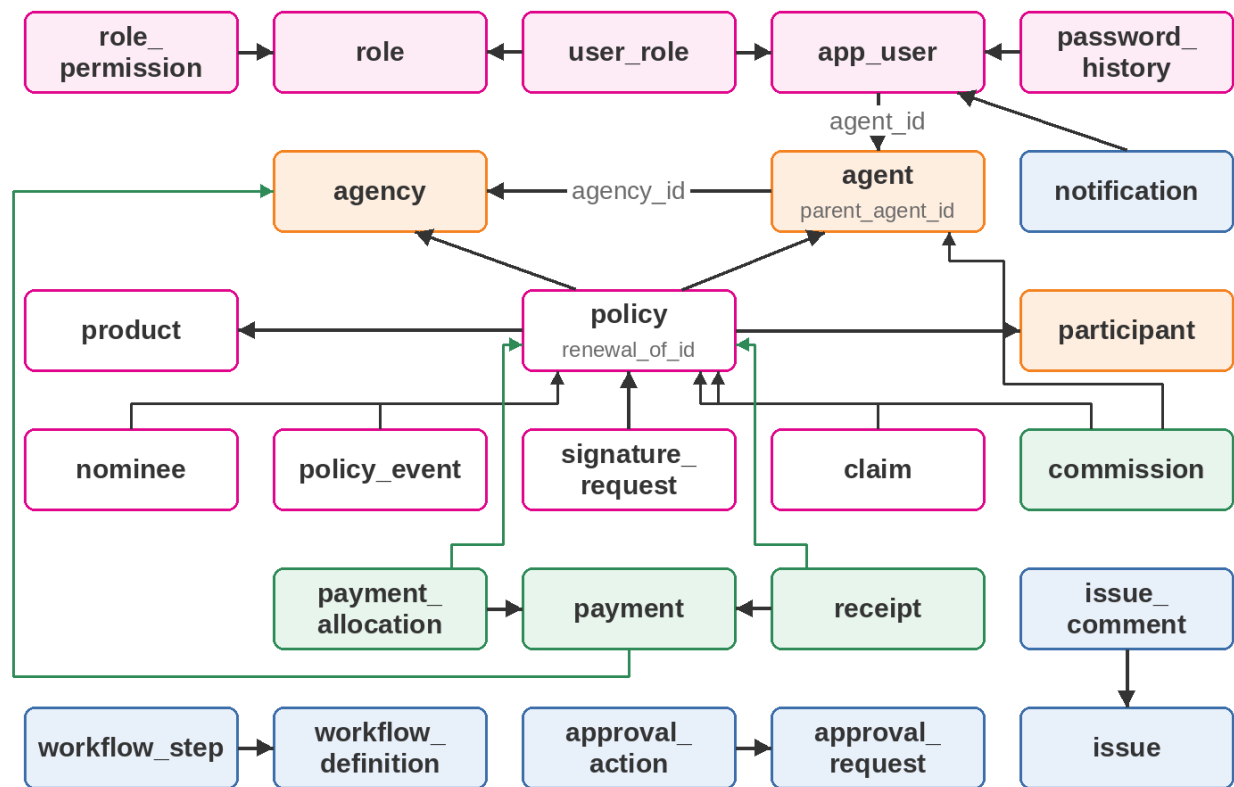
A generator in docs/technical/build reads apps/api/prisma/schema.prisma for tables, columns, keys and relations, and the migration SQL for index names, ON DELETE rules, check constraints, sequences and the audit trigger. It compares every column type and nullability derived from the Prisma schema with the CREATE TABLE statement in the migrations and stops if they differ. Business descriptions are maintained next to the generator, and the build also stops if a table, column, enumeration value, sequence or check constraint has no description.

1.3 Conventions

Item	Convention
Names	Tables and columns use snake_case in PostgreSQL (from @@map and @map). The application uses camelCase model and field names; both are listed.
Primary keys	UUID values generated by the application, except audit_log (BIGSERIAL, to give a total order of events), config_parameter (natural key), user_session (session id) and the two link tables (composite keys).
Timestamps	TIMESTAMP(3) without time zone, written in UTC by the application. created_at defaults to CURRENT_TIMESTAMP; updated_at is set by the application on every update.
Business dates	DATE columns (start_date, payment_date, business_date ...) hold calendar dates in Brunei time (UTC+8).
Money	DECIMAL(14,2) in Brunei dollars; rates DECIMAL(6,4); percentages DECIMAL(5,2). Amounts are never stored as floating point.
Optimistic locking	Tables edited by several users (agent, participant, policy) carry a version column that is checked and incremented on update.
Keys column	PK primary key; UQ unique index; IDX non-unique index; (n/m) is the column's position in a composite index.
JSON	JSONB is used for configuration and variable structures (product config, questionnaire answers, audit before/after values, outbox payloads). Each JSON column is validated by the application before it is written.

Table 1: Dictionary conventions

1.4 Model overview



Tables without foreign keys (polymorphic owner, logs or configuration)

document, audit_log, outbox_message, integration_log, reconciliation_run, eod_run, report_schedule, config_parameter, code_item, aml_screening, aml_watchlist_entry, user_session

Arrow: child table points to its parent (foreign key). Grey text: self-reference.

Figure 1: Entity overview: foreign-key relationships

Table	Cols	Description
Identity and access		
app_user	17	Every person who can sign in: agents and bank officers (portal) and IIFT staff (back-office). Holds credentials, lockout state and the optional link to an agent profile.
password_history	4	Hashes of previous passwords, used to stop reuse of the last N passwords (security.password.history_count).
role	8	Configurable roles (BO-04). Each role belongs to one audience, portal or back-office, and is built from permissions in role_permission.
role_permission	2	Permissions granted to a role. Permission codes are defined in the application (src/common/security/permissions.ts) because each one guards code.
user_role	2	Roles assigned to a user. A user's effective permissions are the union of the permissions of roles that match the user's audience.
user_session	3	Server-side HTTP sessions written by connect-pg-simple. The browser holds only the session id in an HttpOnly cookie; deleting a row ends that session immediately.
Agency / Banca and agents		
agency	14	Distribution partners: agencies (agency channel) and banks (banca channel). Carries the automatic issuance block applied when contributions are overdue (AP-42).
agent	25	Agents, sub-agents and bank officers with their hierarchy, licence, status, AML status and authority limit. Identification numbers are encrypted.

Table	Cols	Description
Participants and AML/KYC		
participant	26	Shared participant (certificate holder) master record, individual or corporate. One record per identification number across all agencies (AP-11). Identification numbers are encrypted.
aml_screening	12	One row per AML/KYC screening of an agent or participant, with match details and the Compliance review decision (BO-13 to BO-15).
aml_watchlist_entry	9	Watch-list entries (sanctions, PEP, internal lists) loaded by Compliance and used by the built-in screening.
Products, quotations and policies		
product	14	Takaful products offered through the portal (Appendix 3) with their rating configuration, required documents and questionnaire, all maintainable without code changes (BO-33).
policy	33	Quotations and policies. A row starts as a DRAFT quotation and keeps the same id through approval, issuance, renewal, endorsement, cancellation and expiry.
nominee	7	Nominees, beneficiaries and executors named on a policy.
policy_event	9	Business history of a policy: every status change and significant action with actor and remarks (AP-31).
signature_request	11	One-time e-signature links e-mailed to participants (AP-62). Only a hash of the token is stored.
Billing		
payment	15	Contribution payments submitted by agencies and banks, with proof of payment, awaiting or after Finance verification (AP-38 to AP-40).
payment_allocation	4	How a payment is split across the policies it pays for (bulk payment, AP-38).
receipt	7	Official e-Receipts issued per policy when a payment is verified (AP-41).
commission	11	Commission or referral fee accrued per agent and policy at issuance, and marked paid when Finance confirms payment (INT-05).
Claims		
claim	14	Claim notifications submitted through the portal (AP-43). Assessment happens in the core system.
Documents		
document	17	Metadata of every uploaded or system-generated document. File content is held encrypted in the document store and addressed by storage_key.
Workflow		
workflow_definition	6	Approval workflow per transaction type (COM-04, BO-16). Inactive definitions approve requests automatically.
workflow_step	6	Levels of a workflow: permission required to approve the level and optional amount threshold above which the level applies (authority limits).
approval_request	18	Maker-checker requests raised for a transaction type, with the proposed change as payload and the approval levels resolved at submission.
approval_action	8	Every action on an approval request: submit, approve, reject or withdraw, with actor and remarks (AP-61).
Issue management		
issue	22	Issues reported from the portal or back-office with priority, assignment and SLA timestamps (AP-55, BO-29 to BO-31).
issue_comment	7	Comments on an issue. Internal comments are visible only to back-office staff.
Notifications		
notification	16	In-app, e-mail and SMS notifications. E-mail and SMS rows are delivered asynchronously by the notification dispatcher with retry.
Audit trail		
audit_log	12	Append-only audit trail of security and business events with before and after values (BO-26 to BO-28, COM-03). A trigger rejects UPDATE and DELETE.

Table	Cols	Description
Configuration and master data		
config_parameter	9	Business parameters administrators can change without a release: password policy, session timeouts, grace period, SLA hours and similar (BO-33, COM-09).
code_item	6	Master data code lists (document types, relationships, nationalities, districts, banks, occupations, claim types, issue categories and others) maintained in the back-office (BO-32).
Integration		
outbox_message	13	Transactional outbox for messages to the core and financial systems. Written in the same transaction as the business change and delivered asynchronously with retry.
integration_log	12	One row per integration call, outbound or inbound, with outcome and duration (INT-11, INT-14).
reconciliation_run	10	Results of daily reconciliation between receipts issued by the portal and receipts posted by the financial system (INT-15).
eod_run	13	End-of-day runs per business date with totals and links to the EOD report and FIN interface file.
report_schedule	12	Scheduled reports e-mailed to recipients daily, weekly or monthly (BO-25).

Table 2: Tables by area

2. Tables

Each table lists its columns in creation order. Foreign key shows the referenced table and column and the ON DELETE rule. Protection marks encrypted, blind-indexed or hashed values.

2.1 app_user

Every person who can sign in: agents and bank officers (portal) and IIFT staff (back-office). Holds credentials, logout state and the optional link to an agent profile. Model User; area Identity and access; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
username	VARCHAR(64)	No		UQ	Sign-in name, stored in lower case and unique across all users.
email	VARCHAR(254)	No			E-mail address for notifications and password reset communication.
full_name	VARCHAR(150)	No			Name shown on screens, in approval history and in the audit trail.
mobile	VARCHAR(20)	Yes			Mobile number for SMS notifications, in international format.
user_type	UserType	No		IDX (1/2)	AGENT, BANCA or STAFF. STAFF users belong to the back-office audience; the others to the portal.
status	UserStatus	No	'ACTIVE'	IDX (2/2)	ACTIVE, LOCKED (temporarily, after failed sign-ins) or DISABLED (by an administrator or core-system status sync).
auth_source	AuthSource	No	'LOCAL'		LOCAL for passwords held by the application; DIRECTORY for back-office users authenticated against Active Directory/LDAP.
password_hash	TEXT	Yes			Argon2id hash of the password (memory 19 MiB, 2 iterations). Null for directory accounts. One-way hash (Argon2id) .
password_changed_at	TIMESTAMP(3)	Yes			When the password was last changed; drives password expiry.
must_change_password	BOOLEAN	No	true		True when the user must change the password at next sign-in (new account or administrator reset).
failed_login_count	INTEGER	No	0		Consecutive failed sign-in attempts since the last success or lock.
locked_until	TIMESTAMP(3)	Yes			End of the current logout period; null when not locked.
last_login_at	TIMESTAMP(3)	Yes			Time of the last successful sign-in.
agent_id	UUID	Yes		UQ FK agent(id); on delete set null	Agent profile of a portal user; null for back-office staff. One user per agent.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).
updated_at	TIMESTAMP(3)	No	Set by the application on every update		Date and time the row was last changed (UTC), maintained by the application.

2.2 password_history

Hashes of previous passwords, used to stop reuse of the last N passwords (security.password.history_count). Model PasswordHistory; area Identity and access; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
user_id	UUID	No		IDX (1/2) FK app_user(id); on delete cascade	User the historical password belonged to.
password_hash	TEXT	No			Argon2id hash of a previous password. One-way hash (Argon2id).
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP	IDX (2/2)	Date and time the row was created (UTC).

2.3 role

Configurable roles (BO-04). Each role belongs to one audience, portal or back-office, and is built from permissions in role_permission. Model Role; area Identity and access; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
code	VARCHAR(50)	No		UQ	Stable role code used by seed data and integrations, e.g. OPERATIONS_SUPERVISOR.
name	VARCHAR(100)	No			Role name shown to administrators.
description	VARCHAR(500)	Yes			What the role is for.
audience	Audience	No			PORTAL or BACKOFFICE. Only permissions of the same audience can be granted to the role.
is_system	BOOLEAN	No	false		True for roles created by the seed script. System roles can be edited but not deleted.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).
updated_at	TIMESTAMP(3)	No	Set by the application on every update		Date and time the row was last changed (UTC), maintained by the application.

2.4 role_permission

Permissions granted to a role. Permission codes are defined in the application (src/common/security/permissions.ts) because each one guards code. Model RolePermission; area Identity and access; primary key role_id, permission.

Column	Data type	Null	Default	Keys / foreign key	Description
role_id	UUID	No		PK FK role(id); on delete cascade	Role receiving the permission.
permission	VARCHAR(80)	No		PK	Permission code, e.g. bo.approve.policies, from the application catalogue.

2.5 user_role

Roles assigned to a user. A user's effective permissions are the union of the permissions of roles that match the user's audience. Model UserRole; area Identity and access; primary key user_id, role_id.

Column	Data type	Null	Default	Keys / foreign key	Description
user_id	UUID	No		PK FK app_user(id); on delete cascade	User holding the role.
role_id	UUID	No		PK FK role(id); on delete cascade	Role assigned to the user.

2.6 user_session

Server-side HTTP sessions written by connect-pg-simple. The browser holds only the session id in an HttpOnly cookie; deleting a row ends that session immediately. Model UserSession; area Identity and access; primary key sid.

Column	Data type	Null	Default	Keys / foreign key	Description
sid	VARCHAR	No		PK	Session identifier carried in the session cookie. Regenerated at sign-in.
sess	JSON	No			Session data: signed-in user snapshot (identity, audience, permissions, agency link), CSRF token and authentication time. Holds the CSRF token; never exposed to the browser.
expire	TIMESTAMP(6)	No		IDX	When the session expires if idle. Expired rows are pruned every 15 minutes.

2.7 agency

Distribution partners: agencies (agency channel) and banks (banca channel). Carries the automatic issuance block applied when contributions are overdue (AP-42). Model Agency; area Agency / Banca and agents; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
code	VARCHAR(20)	No		UQ	Unique agency or bank code used on reports and the FIN interface file.
name	VARCHAR(150)	No			Registered name of the agency or bank.
channel	Channel	No			AGENCY or BANCA distribution channel.
registration_no	VARCHAR(50)	Yes			Business registration number of the agency or bank.
email	VARCHAR(254)	Yes			Main contact e-mail address.
phone	VARCHAR(20)	Yes			Main contact telephone number.
address	VARCHAR(300)	Yes			Business address.
status	AgencyStatus	No	'ACTIVE'		ACTIVE, INACTIVE or SUSPENDED. Users of a non-active agency cannot sign in.
issuance_blocked	BOOLEAN	No	false		True while new business is blocked because an issued policy is unpaid beyond the grace period (AP-42).
issuance_blocked_at	TIMESTAMP(3)	Yes			When the current block was applied.
issuance_block_reason	VARCHAR(300)	Yes			Reason shown to the agency, including the number of overdue policies and the amount.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).
updated_at	TIMESTAMP(3)	No	Set by the application on every update		Date and time the row was last changed (UTC), maintained by the application.

2.8 agent

Agents, sub-agents and bank officers with their hierarchy, licence, status, AML status and authority limit. Identification numbers are encrypted. Model Agent; area Agency / Banca and agents; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
agent_code	VARCHAR(20)	No		UQ	Unique agent code, AG-nnnnnn for agency agents and BK-nnnnnn for bank officers, issued from seq_agent_code.

Column	Data type	Null	Default	Keys / foreign key	Description
agency_id	UUID	No		IDX (1/2) FK agency(id); on delete restrict	Agency or bank the agent belongs to.
agent_type	AgentType	No			MAIN_AGENT, SUB_AGENT or BANKER.
parent_agent_id	UUID	Yes		IDX FK agent(id); on delete set null	Main agent a sub-agent reports to; null for main agents and bank officers without a supervisor.
full_name	VARCHAR(150)	No			Agent's full name as on the identity document.
id_type	IdType	No		UQ (1/2)	Type of identification: NRIC or PASSPORT.
id_number_enc	TEXT	No			Identification number encrypted with AES-256-GCM (format v1:<base64 iv[tag ciphertext>). Decrypted only for authorised display, and then masked. Encrypted (AES-256-GCM, FIELD_ENCRYPTION_KEY) .
id_number_hash	VARCHAR(64)	No		UQ (2/2)	Blind index: HMAC-SHA256 of the normalised identification number. Allows exact search and the uniqueness check without decrypting. Blind index (HMAC-SHA256, FIELD_HASH_KEY) .
date_of_birth	DATE	Yes			Date of birth.
email	VARCHAR(254)	No			Agent's e-mail address for notifications.
mobile	VARCHAR(20)	No			Agent's mobile number for SMS notifications.
address	VARCHAR(300)	Yes			Correspondence address.
branch_name	VARCHAR(100)	Yes			Bank branch of a bank officer.
licence_no	VARCHAR(50)	Yes			Agent licence or registration number.
licence_expiry	DATE	Yes			Licence expiry date, used for expiry warnings.
status	AgentStatus	No	'PENDING'	IDX (2/2)	Registration status: PENDING approval, ACTIVE, INACTIVE, SUSPENDED, TERMINATED or REJECTED.
status_reason	VARCHAR(300)	Yes			Reason for the latest status change, including changes received from the core system.
aml_status	AmlStatus	No	'NOT_SCREENED'		Latest AML/KYC outcome: NOT_SCREENED, CLEAR, FLAGGED or REJECTED.
authority_limit	DECIMAL(14,2)	Yes			Highest sum covered (B\$) the agent may submit without referral; above it the quotation is referred for approval (AP-60). Null means no personal limit.
activated_at	TIMESTAMP(3)	Yes			When the registration was approved and the agent activated.
terminated_at	TIMESTAMP(3)	Yes			When the agent was terminated.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).
updated_at	TIMESTAMP(3)	No	Set by the application on every update		Date and time the row was last changed (UTC), maintained by the application.

Column	Data type	Null	Default	Keys / foreign key	Description
version	INTEGER	No	0		Optimistic-lock counter. Incremented on every update; an update that quotes a stale version is rejected with STALE_RECORD so two users cannot overwrite each other's changes.

2.9 participant

Shared participant (certificate holder) master record, individual or corporate. One record per identification number across all agencies (AP-11). Identification numbers are encrypted. Model Participant; area Participants and AML/KYC; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
participant_no	VARCHAR(20)	No		UQ	Business reference PT/yy/nnnnnn issued from seq_participant_no.
type	ParticipantType	No			INDIVIDUAL or CORPORATE.
full_name	VARCHAR(150)	No		IDX	Full name of the individual or registered name of the company.
id_type	IdType	No		UQ (1/2)	NRIC, PASSPORT or BUSINESS_REG.
id_number_enc	TEXT	No			Identification number encrypted with AES-256-GCM; displayed masked. Encrypted (AES-256-GCM, FIELD_ENCRYPTION_KEY).
id_number_hash	VARCHAR(64)	No		UQ (2/2)	Blind index (HMAC-SHA256) of the normalised identification number; enforces one participant per identification number and supports exact search. Blind index (HMAC-SHA256, FIELD_HASH_KEY).
date_of_birth	DATE	Yes			Date of birth; drives age-next-birthday rating and eligibility.
gender	VARCHAR(10)	Yes			Gender code from master data.
nationality	VARCHAR(50)	Yes			Nationality code from master data; used by eligibility rules (e.g. OSA).
occupation	VARCHAR(100)	Yes			Occupation code from master data.
occupation_class	INTEGER	Yes			Occupational class 1 to 4; the Professional plan accepts class I only.
email	VARCHAR(254)	Yes			Participant e-mail for documents and e-signature links.
mobile	VARCHAR(20)	No			Participant mobile number.
address_line1	VARCHAR(200)	No			First address line.
address_line2	VARCHAR(200)	Yes			Second address line.
postcode	VARCHAR(10)	Yes			Postcode.
district	VARCHAR(50)	Yes			District code from master data.

Column	Data type	Null	Default	Keys / foreign key	Description
contact_person	VARCHAR(150)	Yes			Contact person for a corporate participant.
aml_status	AmlStatus	No	'NOT_SCREENED'		Latest AML/KYC outcome. FLAGGED or REJECTED blocks submission of applications.
aml_screened_at	TIMESTAMP(3)	Yes			When the participant was last screened.
created_by_agent_id	UUID	Yes			Agent who registered the participant (no foreign key, kept for history).
created_by_agency_id	UUID	Yes		IDX	Agency of the registering agent; used for data scoping.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).
updated_at	TIMESTAMP(3)	No	Set by the application on every update		Date and time the row was last changed (UTC), maintained by the application.
version	INTEGER	No	0		Optimistic-lock counter. Incremented on every update; an update that quotes a stale version is rejected with STALE_RECORD so two users cannot overwrite each other's changes.

2.10 aml_screening

One row per AML/KYC screening of an agent or participant, with match details and the Compliance review decision (BO-13 to BO-15). Model AmlScreening; area Participants and AML/KYC; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
subject_type	VARCHAR(20)	No		IDX (1/2)	AGENT or PARTICIPANT.
subject_id	UUID	No		IDX (2/2)	Id of the screened agent or participant.
subject_name	VARCHAR(150)	No			Name screened, kept as at the time of screening.
provider	VARCHAR(50)	No			Source of the result: WATCHLIST, WATCHLIST+EXTERNAL or WATCHLIST+EXTERNAL_UNAVAILABLE.
score	INTEGER	No			Highest match score, 0 to 100. At or above aml.match_threshold the case is flagged.
matches	JSONB	No			Up to ten best matches: list name, matched name, score, reference and reason (NAME or ID_NUMBER).
status	AmlCaseStatus	No		IDX	AUTO_CLEARED, PENDING_REVIEW, CLEARED or CONFIRMED_MATCH.
reviewed_by_id	UUID	Yes			Compliance officer who reviewed the case.
reviewed_at	TIMESTAMP(3)	Yes			When the case was reviewed.
review_remarks	VARCHAR(1000)	Yes			Mandatory reviewer remarks explaining the decision.

Column	Data type	Null	Default	Keys / foreign key	Description
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).

2.11 aml_watchlist_entry

Watch-list entries (sanctions, PEP, internal lists) loaded by Compliance and used by the built-in screening. Model AmlWatchlistEntry; area Participants and AML/KYC; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
list_name	VARCHAR(50)	No			Name of the list the entry came from, e.g. UN, LOCAL, INTERNAL.
full_name	VARCHAR(150)	No			Listed name as published.
normalised	VARCHAR(150)	No		IDX	Name in upper case without punctuation or honorifics; used for candidate selection and similarity scoring.
id_number	VARCHAR(50)	Yes			Identification number of the listed person, where published. An exact match scores 100.
country	VARCHAR(50)	Yes			Country associated with the entry.
reference	VARCHAR(100)	Yes			Reference of the entry in the source list.
active	BOOLEAN	No	true		False when the entry was withdrawn or replaced by a newer import of the list.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).

2.12 product

Takaful products offered through the portal (Appendix 3) with their rating configuration, required documents and questionnaire, all maintainable without code changes (BO-33). Model Product; area Products, quotations and policies; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
code	VARCHAR(20)	No		UQ	Product code, e.g. FTP-HP, PRO, KHR. Prefixes the policy number.
name	VARCHAR(150)	No			Product name shown in the catalogue and on documents.
line_of_business	VARCHAR(50)	No			Line of business, e.g. Mortgage Takaful or Annual.
description	VARCHAR(1000)	No			Short description shown in the product catalogue.

Column	Data type	Null	Default	Keys / foreign key	Description
rating_engine	VARCHAR(30)	No			Rating engine that interprets config: FINANCING (decreasing-term financing products) or FIXED_PLAN (plan-based annual products).
config	JSONB	No			Rating configuration: rate tables, plans, loadings, limits, eligibility, wakalah fee and commission rates. Validated by the rating engine before it is saved.
required_documents	JSONB	No			Documents required for the product: type, label and whether mandatory before submission.
questionnaire	JSONB	No			Health or risk questions; a 'yes' to a question marked referIfYes refers the application for approval.
payment_before_issuance	BOOLEAN	No	true		True for pay-first products; false for issue-then-pay products, which must be paid within the grace period.
allow_renewal	BOOLEAN	No	true		Whether policies of the product can be renewed from the portal.
active	BOOLEAN	No	true		Inactive products are hidden from the catalogue and cannot be quoted.
sort_order	INTEGER	No	0		Display order in the catalogue.
updated_at	TIMESTAMP(3)	No	Set by the application on every update		Date and time the row was last changed (UTC), maintained by the application.

2.13 policy

Quotations and policies. A row starts as a DRAFT quotation and keeps the same id through approval, issuance, renewal, endorsement, cancellation and expiry. Model Policy; area Products, quotations and policies; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
quotation_no	VARCHAR(20)	No		UQ	Quotation reference QT/yy/nnnnnn issued from seq_quotation_no when the draft is created.
policy_no	VARCHAR(20)	Yes		UQ	Policy number <product code>/yy/nnnnnn issued from seq_policy_no at issuance; null before issuance.
product_id	UUID	No		FK product(id); on delete restrict	Product quoted.
participant_id	UUID	No		IDX FK participant(id); on delete restrict	Participant (certificate holder).
agent_id	UUID	No		IDX (1/2) FK agent(id); on delete restrict	Agent or bank officer who owns the business.
agency_id	UUID	No		IDX (1/2) FK agency(id); on delete restrict	Agency or bank of the agent; used for data scoping and the grace-period rule.

Column	Data type	Null	Default	Keys / foreign key	Description
status	PolicyStatus	No	'DRAFT'	IDX (2/2), IDX (1/3)	Lifecycle status: DRAFT, PENDING_APPROVAL, PENDING_PAYMENT, ACTIVE, REJECTED, EXPIRED or CANCELLED.
payment_status	PolicyPaymentStatus	No	'UNPAID'	IDX (2/3)	UNPAID, PENDING_VERIFICATION (payment submitted) or PAID.
plan_code	VARCHAR(20)	Yes			Selected plan for plan-based products, e.g. A, B, C.
coverage_type	VARCHAR(20)	Yes			Coverage option, e.g. INDIVIDUAL or WIDER for Khairat.
sum_covered	DECIMAL(14,2)	No			Sum covered in B\$.
term_months	INTEGER	No			Cover term in months.
start_date	DATE	Yes			Cover start date.
end_date	DATE	Yes			Cover end date; the expiry job sets status EXPIRED after this date.
contribution	DECIMAL(14,2)	No			Total contribution payable in B\$, including any additional cover.
contribution_breakdown	JSONB	No			Contribution lines, tabarru' and wakalah fee split and commission rate, as rated.
risk_details	JSONB	No			Product-specific rating inputs, e.g. financing amount, tenure and profit rate, employer details, student institution.
questionnaire	JSONB	Yes			Answers to the product questionnaire.
referral_reasons	TEXT[]	Yes	Empty array		Reasons the application was referred for approval (authority limit, high-risk limit, questionnaire, quality check). Empty when not referred.
outstanding_amount	DECIMAL(14,2)	No	0		Contribution not yet paid in B\$.
payment_due_date	DATE	Yes		IDX (3/3)	Last day to submit payment for issue-then-pay products (issuance date plus grace days).
submitted_at	TIMESTAMP(3)	Yes			When the quotation was last submitted.
approved_at	TIMESTAMP(3)	Yes			When the referral was approved.
issued_at	TIMESTAMP(3)	Yes		IDX	When the policy was issued; drives the EOD selection.
rejected_reason	VARCHAR(1000)	Yes			Reason given by the checker when the application was rejected (AP-51).
cancelled_at	TIMESTAMP(3)	Yes			When the policy was cancelled.
cancellation_reason	VARCHAR(1000)	Yes			Approved cancellation reason.
renewal_of_id	UUID	Yes		FK policy(id); on delete set null	Policy this one renews; null for new business.
created_by_id	UUID	No			User who created the quotation.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).
updated_at	TIMESTAMP(3)	No	Set by the application on every update		Date and time the row was last changed (UTC), maintained by the application.

Column	Data type	Null	Default	Keys / foreign key	Description
version	INTEGER	No	0		Optimistic-lock counter. Incremented on every update; an update that quotes a stale version is rejected with STALE_RECORD so two users cannot overwrite each other's changes.

2.14 nominee

Nominees, beneficiaries and executors named on a policy. Model Nominee; area Products, quotations and policies; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
policy_id	UUID	No		IDX FK policy(id); on delete cascade	Policy the nominee belongs to.
full_name	VARCHAR(150)	No			Nominee's full name.
id_number_enc	TEXT	Yes			Nominee's identification number encrypted with AES-256-GCM; optional. Encrypted (AES-256-GCM, FIELD_ENCRYPTION_KEY).
relationship	VARCHAR(50)	No			Relationship to the participant, from master data.
role	NomineeRole	No	'NOMINEE'		NOMINEE, BENEFICIARY or EXECUTOR.
share_percent	DECIMAL(5,2)	No			Share of the benefit in per cent, greater than 0 and at most 100; shares on a policy total 100.

2.15 policy_event

Business history of a policy: every status change and significant action with actor and remarks (AP-31). Model PolicyEvent; area Products, quotations and policies; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
policy_id	UUID	No		IDX (1/2) FK policy(id); on delete cascade	Policy the event belongs to.
action	VARCHAR(50)	No			Event code, e.g. QUOTATION_CREATED, SUBMITTED, REFERRED, ISSUED, PARTICIPANT_SIGNED, PAYMENT_APPLIED, EXPIRED.
from_status	PolicyStatus	Yes			Status before the event, when the event changed status.
to_status	PolicyStatus	Yes			Status after the event.
remarks	VARCHAR(1000)	Yes			Free-text detail, e.g. referral reasons or rejection remarks.

Column	Data type	Null	Default	Keys / foreign key	Description
actor_id	UUID	Yes			User who performed the action; null for system jobs and participants.
actor_name	VARCHAR(150)	Yes			Name of the actor at the time: user name, participant name or System.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP	IDX (2/2)	Date and time the row was created (UTC).

2.16 signature_request

One-time e-signature links e-mailed to participants (AP-62). Only a hash of the token is stored. Model SignatureRequest; area Products, quotations and policies; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
policy_id	UUID	No		IDX FK policy(id); on delete cascade	Quotation to be signed.
token_hash	VARCHAR(64)	No		UQ	SHA-256 of the one-time token in the e-mailed link. The token itself is never stored. One-way hash (SHA-256 of one-time token).
recipient_name	VARCHAR(150)	No			Participant name the signer must type to confirm identity.
recipient_email	VARCHAR(254)	No			Address the link was sent to.
expires_at	TIMESTAMP(3)	No			Link expiry (policy.esign_link_hours after sending).
signed_at	TIMESTAMP(3)	Yes			When the participant signed; a non-null value makes the link unusable.
signer_ip	VARCHAR(64)	Yes			IP address the signature was made from.
document_id	UUID	Yes			Signature image stored as a document.
created_by_id	UUID	No			Agent user who sent the link.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).

2.17 payment

Contribution payments submitted by agencies and banks, with proof of payment, awaiting or after Finance verification (AP-38 to AP-40). Model Payment; area Billing; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
payment_no	VARCHAR(20)	No		UQ	Payment reference PY/yy/nnnnnn issued from seq_payment_no.

Column	Data type	Null	Default	Keys / foreign key	Description
agency_id	UUID	No		IDX (1/2) FK agency(id); on delete restrict	Agency or bank that made the payment.
submitted_by_id	UUID	No			User who submitted the payment.
method	PaymentMethod	No			BANK_TRANSFER, CHEQUE, CASH_DEPOSIT or ONLINE.
bank_name	VARCHAR(100)	Yes			Paying bank, from master data.
reference_no	VARCHAR(50)	No			Bank transaction or cheque reference; carried to the FIN interface file.
payment_date	DATE	No			Date the payment was made.
total_amount	DECIMAL(14,2)	No			Total paid in B\$; must be greater than zero and equal the sum of allocations.
status	PaymentStatus	No	'PENDING_VERIFICATION'	IDX (2/2), IDX (1/2)	PENDING_VERIFICATION, VERIFIED or REJECTED.
remarks	VARCHAR(500)	Yes			Remarks from the submitter.
verified_by_id	UUID	Yes			Finance officer who verified or rejected the payment.
verified_at	TIMESTAMP(3)	Yes			When the payment was verified or rejected.
rejection_reason	VARCHAR(500)	Yes			Reason the payment was rejected.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP	IDX (2/2)	Date and time the row was created (UTC).

2.18 payment_allocation

How a payment is split across the policies it pays for (bulk payment, AP-38). Model PaymentAllocation; area Billing; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
payment_id	UUID	No		UQ (1/2) FK payment(id); on delete cascade	Payment being allocated.
policy_id	UUID	No		IDX, UQ (2/2) FK policy(id); on delete restrict	Policy the amount pays for.
amount	DECIMAL(14,2)	No			Amount allocated to the policy in B\$; greater than zero.

2.19 receipt

Official e-Receipts issued per policy when a payment is verified (AP-41). Model Receipt; area Billing; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
receipt_no	VARCHAR(20)	No		UQ	Receipt number RC/yy/nnnnnn issued from seq_receipt_no; reconciled with the financial system.
payment_id	UUID	No		FK payment(id); on delete restrict	Verified payment the receipt was issued from.
policy_id	UUID	No		IDX FK policy(id); on delete restrict	Policy the receipt is for.
amount	DECIMAL(14,2)	No			Amount received for the policy in B\$.
issued_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP	IDX	When the receipt was issued; drives the EOD and reconciliation selection.
document_id	UUID	Yes			PDF e-Receipt stored as a document.

2.20 commission

Commission or referral fee accrued per agent and policy at issuance, and marked paid when Finance confirms payment (INT-05). Model Commission; area Billing; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
agent_id	UUID	No		IDX (1/2), UQ (2/2) FK agent(id); on delete restrict	Agent earning the commission or referral fee.
policy_id	UUID	No		UQ (1/2) FK policy(id); on delete restrict	Policy the commission relates to.
period	VARCHAR(7)	No		IDX (2/2)	Accrual period as YYYY-MM.
contribution	DECIMAL(14,2)	No			Contribution the commission was calculated on, in B\$.
rate	DECIMAL(6,4)	No			Commission rate applied, as a fraction (0.1000 = 10%).
amount	DECIMAL(14,2)	No			Commission amount in B\$.
status	CommissionStatus	No	'ACCRUED'		ACCRUED until Finance confirms payment, then PAID.
paid_at	TIMESTAMP(3)	Yes			Payment date confirmed by Finance through the inbound API.
source	VARCHAR(20)	No			Where the accrual was created; PORTAL for accruals at issuance.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).

2.21 claim

Claim notifications submitted through the portal (AP-43). Assessment happens in the core system. Model Claim; area Claims; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
claim_no	VARCHAR(20)	No		UQ	Claim notification reference CL/yy/nnnnnn issued from seq_claim_no.
policy_id	UUID	No		IDX FK policy(id); on delete restrict	Policy the claim is notified under.
agent_id	UUID	No			Agent who notified the claim (copied from the policy for data scoping).
agency_id	UUID	No		IDX (1/2)	Agency of the agent (copied from the policy for data scoping).
claim_type	VARCHAR(50)	No			Claim type code from master data: DEATH, TPD, ACCIDENT, MEDICAL, REPATRIATION or OTHER.
event_date	DATE	No			Date of the insured event.
description	VARCHAR(2000)	No			Description of the event.
claimed_amount	DECIMAL(14,2)	Yes			Amount claimed in B\$, where known.
status	ClaimStatus	No	'SUBMITTED'	IDX (2/2)	SUBMITTED, UNDER_REVIEW, ACKNOWLEDGED, REJECTED or CLOSED.
remarks	VARCHAR(1000)	Yes			Back-office remarks visible to the agent.
submitted_by_id	UUID	No			User who submitted the notification.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).
updated_at	TIMESTAMP(3)	No	Set by the application on every update		Date and time the row was last changed (UTC), maintained by the application.

2.22 document

Metadata of every uploaded or system-generated document. File content is held encrypted in the document store and addressed by storage_key. Model Document; area Documents; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
owner_type	DocumentOwnerType	No		IDX (1/2)	Type of record the document belongs to: AGENT, AGENCY, PARTICIPANT, POLICY, PAYMENT, CLAIM, ISSUE or REPORT.

Column	Data type	Null	Default	Keys / foreign key	Description
owner_id	UUID	No		IDX (2/2)	Id of the owning record (polymorphic, no foreign key).
doc_type	VARCHAR(50)	No			Document type code from master data, e.g. IC_COPY, PROPOSAL_FORM, PAYMENT_PROOF, POLICY_SCHEDULE, RECEIPT, EOD_REPORT, FIN_FILE.
file_name	VARCHAR(255)	No			Sanitised file name; the extension always matches the detected content type.
mime_type	VARCHAR(100)	No			Content type detected from the file's magic bytes (PDF, PNG, JPEG) or set for generated files.
size_bytes	INTEGER	No			File size in bytes before encryption.
storage_key	VARCHAR(255)	No		UQ	Location in the encrypted store, yyyy/mm/<uuid>.bin. Generated by the application; never derived from user input.
sha256	VARCHAR(64)	No			SHA-256 checksum of the original content, for integrity and duplicate checks.
status	DocumentStatus	No	'UPLOADED'	IDX	UPLOADED, VERIFIED or REJECTED by a back-office checker.
expiry_date	DATE	Yes			Expiry date of the document (e.g. passport, licence) for expiry warnings.
remarks	VARCHAR(500)	Yes			Verifier remarks, required on rejection.
system_generated	BOOLEAN	No	false		True for documents produced by the system (schedule, receipt, reports).
uploaded_by_id	UUID	Yes			User who uploaded the file; null for system-generated files and participant e-signatures.
verified_by_id	UUID	Yes			Back-office user who verified or rejected the document.
verified_at	TIMESTAMP(3)	Yes			When the document was verified or rejected.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).

2.23 workflow_definition

Approval workflow per transaction type (COM-04, BO-16). Inactive definitions approve requests automatically. Model WorkflowDefinition; area Workflow; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
type	ApprovalType	No		UQ	Transaction type governed by the workflow; one definition per type.
name	VARCHAR(100)	No			Workflow name shown to administrators.
description	VARCHAR(500)	Yes			What the workflow covers.
active	BOOLEAN	No	true		When false, requests of this type are approved automatically on submission.

Column	Data type	Null	Default	Keys / foreign key	Description
updated_at	TIMESTAMP(3)	No	Set by the application on every update		Date and time the row was last changed (UTC), maintained by the application.

2.24 workflow_step

Levels of a workflow: permission required to approve the level and optional amount threshold above which the level applies (authority limits). Model WorkflowStep; area Workflow; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
definition_id	UUID	No		UQ (1/2) FK workflow_definition(id); on delete cascade	Workflow the step belongs to.
level	INTEGER	No		UQ (2/2)	Approval level, starting at 1.
name	VARCHAR(100)	No			Step name shown in the inbox and approval history.
permission	VARCHAR(80)	No			Permission a checker needs to approve this level.
min_amount	DECIMAL(14,2)	Yes			Amount threshold in B\$; the level applies only when the request amount is at or above it. Null means always.

2.25 approval_request

Maker-checker requests raised for a transaction type, with the proposed change as payload and the approval levels resolved at submission. Model ApprovalRequest; area Workflow; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
request_no	VARCHAR(20)	No		UQ	Request reference RQ/yy/nnnnnn issued from seq_request_no.
type	ApprovalType	No		IDX (2/2)	Transaction type, e.g. AGENT_REGISTRATION, POLICY_REFERRAL, PAYMENT_VERIFICATION.
entity_type	VARCHAR(30)	No		IDX (1/2)	Type of record the request is about, e.g. Agent, Policy, Payment.
entity_id	UUID	No		IDX (2/2)	Id of the record the request is about.
summary	VARCHAR(300)	No			One-line summary shown in the inbox.

Column	Data type	Null	Default	Keys / foreign key	Description
payload	JSONB	No			Proposed change or context for the checker, e.g. before and after values of a profile update.
amount	DECIMAL(14,2)	Yes			Amount used to select levels with thresholds, e.g. sum covered.
status	ApprovalStatus	No	'PENDING'	IDX (1/2)	PENDING, APPROVED, REJECTED or WITHDRAWN.
current_level	INTEGER	No	1		Level awaiting a decision.
total_levels	INTEGER	No			Number of levels that apply to this request, fixed at submission.
level_permissions	TEXT[]	Yes			Permission required at each level, in order, fixed at submission so a later workflow change does not alter a request in flight.
maker_id	UUID	No		IDX	User who raised the request. Cannot approve it.
maker_name	VARCHAR(150)	No			Maker's name at the time.
agency_id	UUID	Yes			Agency the request relates to, for portal visibility.
submitted_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		When the request was raised.
decided_at	TIMESTAMP(3)	Yes			When the request reached its final status.
final_remarks	VARCHAR(1000)	Yes			Remarks of the final decision.

2.26 approval_action

Every action on an approval request: submit, approve, reject or withdraw, with actor and remarks (AP-61). Model ApprovalAction; area Workflow; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
request_id	UUID	No		IDX FK approval_request(id); on delete cascade	Request the action belongs to.
level	INTEGER	No			Level acted on; 0 for submission.
action	ApprovalActionType	No			SUBMIT, APPROVE, REJECT or WITHDRAW.
actor_id	UUID	No			User who acted.
actor_name	VARCHAR(150)	No			Actor's name at the time.
remarks	VARCHAR(1000)	Yes			Remarks; mandatory for rejection.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).

2.27 issue

Issues reported from the portal or back-office with priority, assignment and SLA timestamps (AP-55, BO-29 to BO-31). Model Issue; area Issue management; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
issue_no	VARCHAR(20)	No		UQ	Issue reference IS/yy/nnnnnn issued from seq_issue_no.
title	VARCHAR(200)	No			Short summary of the issue.
description	VARCHAR(4000)	No			Full description.
category	VARCHAR(50)	No			Issue category code from master data.
priority	IssuePriority	No		IDX (2/2)	CRITICAL, HIGH, MEDIUM or LOW; sets the SLA targets.
status	IssueStatus	No	'OPEN'	IDX (1/2)	OPEN, ASSIGNED, IN_PROGRESS, RESOLVED or CLOSED.
reported_by_id	UUID	No		IDX	User who reported the issue.
reported_by_name	VARCHAR(150)	No			Reporter's name at the time.
agency_id	UUID	Yes			Reporter's agency, for portal users.
assigned_to_id	UUID	Yes		IDX	Support user the issue is assigned to.
assigned_to_name	VARCHAR(150)	Yes			Assignee's name at the time.
assigned_team	VARCHAR(50)	Yes			Support team the issue is assigned to.
response_due_at	TIMESTAMP(3)	No			Response target: creation time plus the response hours for the priority.
resolution_due_at	TIMESTAMP(3)	No			Resolution target: creation time plus the resolution hours for the priority.
first_responded_at	TIMESTAMP(3)	Yes			When support first responded.
resolved_at	TIMESTAMP(3)	Yes			When the issue was resolved.
closed_at	TIMESTAMP(3)	Yes			When the issue was closed.
sla_breached	BOOLEAN	No	false		Set by the SLA monitor when a target is missed; support is alerted once.
resolution	VARCHAR(2000)	Yes			Resolution notes.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).
updated_at	TIMESTAMP(3)	No	Set by the application on every update		Date and time the row was last changed (UTC), maintained by the application.

2.28 issue_comment

Comments on an issue. Internal comments are visible only to back-office staff. Model IssueComment; area Issue management; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
issue_id	UUID	No		IDX FK issue(id); on delete cascade	Issue the comment belongs to.
author_id	UUID	No			User who wrote the comment.
author_name	VARCHAR(150)	No			Author's name at the time.
body	VARCHAR(4000)	No			Comment text.
internal	BOOLEAN	No	false		True for back-office-only comments.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).

2.29 notification

In-app, e-mail and SMS notifications. E-mail and SMS rows are delivered asynchronously by the notification dispatcher with retry. Model Notification; area Notifications; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
user_id	UUID	Yes		IDX (1/3) FK app_user(id); on delete cascade	Recipient user; null for external recipients such as participants and report subscribers.
channel	NotificationChannel	No		IDX (2/3), IDX (2/2)	IN_APP, EMAIL or SMS.
recipient	VARCHAR(254)	No			E-mail address, mobile number or user id, depending on the channel.
event_type	VARCHAR(50)	No			Event that caused the notification, e.g. POLICY_ISSUED, APPROVAL_REQUIRED, AGENCY_BLOCKED.
subject	VARCHAR(200)	No			Subject line or in-app title.
body	VARCHAR(4000)	No			Message text. For sensitive messages (e.g. e-signature links) replaced with a placeholder after delivery. Removed after delivery when sensitive = true.
link	VARCHAR(300)	Yes			Application path the in-app notification opens.
attachment_ids	TEXT[]	Yes	Empty array		Documents attached to the e-mail.
sensitive	BOOLEAN	No	false		True when the body must be removed once delivered.

Column	Data type	Null	Default	Keys / foreign key	Description
status	DeliveryStatus	No	'PENDING'	IDX (1/2)	PENDING, SENT or FAILED (after notification.max_attempts attempts).
attempts	INTEGER	No	0		Delivery attempts so far.
last_error	VARCHAR(1000)	Yes			Error from the last failed attempt.
read_at	TIMESTAMP(3)	Yes		IDX (3/3)	When the user read an in-app notification.
sent_at	TIMESTAMP(3)	Yes			When the e-mail or SMS was accepted by the gateway.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).

2.30 audit_log

Append-only audit trail of security and business events with before and after values (BO-26 to BO-28, COM-03). A trigger rejects UPDATE and DELETE. Model AuditLog; area Audit trail; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	BIGSERIAL	No	Next value of the column sequence	PK	Sequential primary key; gives a total order of audit events.
occurred_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP	IDX, IDX (2/2)	When the event happened (UTC).
actor_id	UUID	Yes		IDX (1/2)	User who performed the action; null for system jobs and inbound integrations.
actor_name	VARCHAR(150)	Yes			Username of the actor, or 'system'.
action	VARCHAR(60)	No		IDX	Event code, e.g. LOGIN, LOGIN_FAILED, ACCOUNT_LOCKED, APPROVAL_DECIDED, PRODUCT_UPDATED, AGENCY_BLOCKED.
entity_type	VARCHAR(40)	No		IDX (1/2)	Type of record affected.
entity_id	VARCHAR(64)	Yes		IDX (2/2)	Id of the record affected.
before	JSONB	Yes			Values before the change. Passwords, hashes, encrypted identifiers, tokens and storage keys are written as a fixed REDACTED marker.
after	JSONB	Yes			Values after the change, redacted in the same way.
ip_address	VARCHAR(64)	Yes			Client IP address as seen through the reverse proxy.
user_agent	VARCHAR(300)	Yes			Client browser user agent, truncated to 300 characters.
correlation_id	VARCHAR(64)	Yes			Request correlation id (X-Request-Id); links the event to application logs.

2.31 config_parameter

Business parameters administrators can change without a release: password policy, session timeouts, grace period, SLA hours and similar (BO-33, COM-09). Model ConfigParameter; area Configuration and master data; primary key key.

Column	Data type	Null	Default	Keys / foreign key	Description
key	VARCHAR(80)	No		PK	Parameter key, e.g. security.lockout.max_attempts.
value	VARCHAR(1000)	No			Current value as text, interpreted according to value_type.
value_type	ParameterType	No			STRING, INTEGER, DECIMAL or BOOLEAN.
category	VARCHAR(50)	No			Grouping shown in the back-office, e.g. Security, Billing, Issue SLA.
description	VARCHAR(500)	No			What the parameter controls.
min_value	DECIMAL(14,2)	Yes			Lowest value accepted for numeric parameters.
max_value	DECIMAL(14,2)	Yes			Highest value accepted for numeric parameters.
updated_by_id	UUID	Yes			Administrator who last changed the value.
updated_at	TIMESTAMP(3)	No	Set by the application on every update		Date and time the row was last changed (UTC), maintained by the application.

2.32 code_item

Master data code lists (document types, relationships, nationalities, districts, banks, occupations, claim types, issue categories and others) maintained in the back-office (BO-32). Model CodeItem; area Configuration and master data; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
category	VARCHAR(50)	No		UQ (1/2)	Code list name, e.g. DOCUMENT_TYPE, RELATIONSHIP, DISTRICT, BANK, CLAIM_TYPE.
code	VARCHAR(50)	No		UQ (2/2)	Stable code stored on business records.
label	VARCHAR(150)	No			Label shown to users.
active	BOOLEAN	No	true		Inactive codes are hidden from new entries but remain valid on existing records.
sort_order	INTEGER	No	0		Display order within the list.

2.33 outbox_message

Transactional outbox for messages to the core and financial systems. Written in the same transaction as the business change and delivered asynchronously with retry. Model OutboxMessage; area Integration; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Primary key; also sent as the Idempotency-Key header so the receiver can discard duplicates.
system	IntegrationSystem	No		IDX (1/2)	Target system: CORE or FINANCE.
operation	VARCHAR(60)	No			Operation code from the interface specification, e.g. POLICY_ISSUED, RECEIPT_POSTED, EOD_POSTING.
payload	JSONB	No			JSON message body.
status	OutboxStatus	No	'PENDING'	IDX (1/2)	PENDING, SENT, FAILED (retry scheduled) or DEAD (attempts exhausted or not retryable).
attempts	INTEGER	No	0		Delivery attempts so far.
next_attempt_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP	IDX (2/2)	Earliest time of the next attempt (exponential back-off, capped at 60 minutes).
last_error	VARCHAR(1000)	Yes			Error from the last failed attempt.
aggregate_type	VARCHAR(30)	Yes			Type of business record that produced the message, e.g. Policy, Receipt.
aggregate_id	VARCHAR(64)	Yes			Id of the business record.
correlation_id	VARCHAR(64)	Yes			Correlation id of the request that produced the message.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP	IDX (2/2)	Date and time the row was created (UTC).
processed_at	TIMESTAMP(3)	Yes			When the message was delivered.

2.34 integration_log

One row per integration call, outbound or inbound, with outcome and duration (INT-11, INT-14). Model IntegrationLog; area Integration; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
system	IntegrationSystem	No		IDX (1/2)	CORE, FINANCE, AML, EMAIL, SMS or DIRECTORY.
operation	VARCHAR(60)	No			Operation or notification event code.
direction	IntegrationDirection	No			OUTBOUND (portal calls the system) or INBOUND (system calls the portal).
success	BOOLEAN	No		IDX (1/2)	Whether the call succeeded.

Column	Data type	Null	Default	Keys / foreign key	Description
duration_ms	INTEGER	No			Call duration in milliseconds.
reference	VARCHAR(100)	Yes			Reference returned by the receiver, or the business reference of an inbound call. Simulated deliveries are marked '(simulated)'.
request_summary	VARCHAR(1000)	Yes			Short description of the request, e.g. aggregate type and id. Never the full payload.
response_summary	VARCHAR(1000)	Yes			Short description of the response.
error_message	VARCHAR(1000)	Yes			Error text for failed calls.
outbox_id	UUID	Yes			Outbox message delivered by this call, if any.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP	IDX (2/2)	Date and time the row was created (UTC).

2.35 reconciliation_run

Results of daily reconciliation between receipts issued by the portal and receipts posted by the financial system (INT-15). Model ReconciliationRun; area Integration; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
business_date	DATE	No		IDX (1/2)	Business date reconciled.
system	IntegrationSystem	No		IDX (2/2)	System reconciled against; FINANCE.
expected_count	INTEGER	No			Receipts issued by the portal on the date.
matched_count	INTEGER	No			Receipts also reported as posted by the financial system.
expected_amount	DECIMAL(14,2)	No			Total of receipts issued, in B\$.
matched_amount	DECIMAL(14,2)	No			Total of matched receipts, in B\$.
status	ReconciliationStatus	No			MATCHED or MISMATCH.
details	JSONB	No			Unmatched receipt numbers for investigation.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).

2.36 eod_run

End-of-day runs per business date with totals and links to the EOD report and FIN interface file. Model EodRun; area Integration; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
business_date	DATE	No		UQ	Business date processed; one run per date (a re-run updates the same row).
status	JobStatus	No			RUNNING, COMPLETED or FAILED.
started_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		When the run started.
finished_at	TIMESTAMP(3)	Yes			When the run finished.
policies_issued	INTEGER	No	0		Policies issued on the date.
receipts_issued	INTEGER	No	0		Receipts issued on the date.
total_contribution	DECIMAL(14,2)	No	0		Total contribution of policies issued, in B\$.
total_receipts	DECIMAL(14,2)	No	0		Total of receipts issued, in B\$.
report_document_id	UUID	Yes			EOD issuance report (Excel) stored as a document.
fin_file_document_id	UUID	Yes			FIN interface file (CSV) stored as a document.
error_message	VARCHAR(1000)	Yes			Error text of a failed run.
triggered_by	VARCHAR(150)	No			'Scheduler' or the name of the user who ran it from the back-office.

2.37 report_schedule

Scheduled reports e-mailed to recipients daily, weekly or monthly (BO-25). Model ReportSchedule; area Integration; primary key id.

Column	Data type	Null	Default	Keys / foreign key	Description
id	UUID	No	UUID v4 generated by the application	PK	Surrogate primary key.
report_code	VARCHAR(50)	No			Report from the report catalogue, e.g. POLICY_REGISTER, COLLECTIONS.
name	VARCHAR(150)	No			Schedule name used in the e-mail subject.
frequency	ReportFrequency	No			DAILY (07:00), WEEKLY (Monday) or MONTHLY (1st), Brunei time.
format	ExportFormat	No			XLSX, CSV or PDF.
recipients	TEXT[]	Yes			E-mail addresses that receive the report.

Column	Data type	Null	Default	Keys / foreign key	Description
filters	JSONB	No			Report period, e.g. PREVIOUS_DAY, PREVIOUS_7_DAYS, PREVIOUS_MONTH, MONTH_TO_DATE.
active	BOOLEAN	No	true	IDX (1/2)	Paused schedules are not run.
last_run_at	TIMESTAMP(3)	Yes			When the schedule last ran.
next_run_at	TIMESTAMP(3)	No		IDX (2/2)	When the schedule runs next.
created_by_id	UUID	No			User who created the schedule.
created_at	TIMESTAMP(3)	No	CURRENT_TIMESTAMP		Date and time the row was created (UTC).

3. Relationships

Every foreign key in the database. The parent row must exist before a child row can reference it. RESTRICT prevents deleting a parent that still has children; CASCADE deletes dependent rows with the parent; SET NULL clears the reference.

Parent	Child	Child column	Cardinality	On delete	Constraint
agent	app_user	agent_id	1 : 0..1	SET NULL	app_user_agent_id_fkey
app_user	password_history	user_id	1 : 0..n	CASCADE	password_history_user_id_fkey
role	role_permission	role_id	1 : 0..n	CASCADE	role_permission_role_id_fkey
app_user	user_role	user_id	1 : 0..n	CASCADE	user_role_user_id_fkey
role	user_role	role_id	1 : 0..n	CASCADE	user_role_role_id_fkey
agency	agent	agency_id	1 : 0..n	RESTRICT	agent_agency_id_fkey
agent	agent	parent_agent_id	1 : 0..n	SET NULL	agent_parent_agent_id_fkey
product	policy	product_id	1 : 0..n	RESTRICT	policy_product_id_fkey
participant	policy	participant_id	1 : 0..n	RESTRICT	policy_participant_id_fkey
agent	policy	agent_id	1 : 0..n	RESTRICT	policy_agent_id_fkey
agency	policy	agency_id	1 : 0..n	RESTRICT	policy_agency_id_fkey
policy	policy	renewal_of_id	1 : 0..n	SET NULL	policy_renewal_of_id_fkey
policy	nominee	policy_id	1 : 0..n	CASCADE	nominee_policy_id_fkey
policy	policy_event	policy_id	1 : 0..n	CASCADE	policy_event_policy_id_fkey
policy	signature_request	policy_id	1 : 0..n	CASCADE	signature_request_policy_id_fkey
agency	payment	agency_id	1 : 0..n	RESTRICT	payment_agency_id_fkey
payment	payment_allocation	payment_id	1 : 0..n	CASCADE	payment_allocation_payment_id_fkey
policy	payment_allocation	policy_id	1 : 0..n	RESTRICT	payment_allocation_policy_id_fkey
payment	receipt	payment_id	1 : 0..n	RESTRICT	receipt_payment_id_fkey
policy	receipt	policy_id	1 : 0..n	RESTRICT	receipt_policy_id_fkey
agent	commission	agent_id	1 : 0..n	RESTRICT	commission_agent_id_fkey
policy	commission	policy_id	1 : 0..n	RESTRICT	commission_policy_id_fkey
policy	claim	policy_id	1 : 0..n	RESTRICT	claim_policy_id_fkey

Parent	Child	Child column	Cardinality	On delete	Constraint
workflow_definition	workflow_step	definition_id	1 : 0..n	CASCADE	workflow_step_definition_id_fkey
approval_request	approval_action	request_id	1 : 0..n	CASCADE	approval_action_request_id_fkey
issue	issue_comment	issue_id	1 : 0..n	CASCADE	issue_comment_issue_id_fkey
app_user	notification	user_id	1 : 0..n	CASCADE	notification_user_id_fkey

Table 3: Foreign keys

Polymorphic references are deliberately not foreign keys: document.owner_type/owner_id, approval_request.entity_type/entity_id, aml_screening.subject_type/subject_id, outbox_message.aggregate_type/aggregate_id and audit_log.entity_type/entity_id. The application checks these references, and history tables keep the actor name so records remain readable after a user is deactivated.

4. Indexes

Indexes created by the migrations, grouped by table. Composite indexes serve the most frequent list screens: for example policy (agency_id, status) for the portal policy list and outbox_message (status, next_attempt_at) for the dispatcher.

Table	Index	Columns	Unique	Purpose
app_user	app_user_pkey	id	Yes	Primary key
app_user	app_user_agent_id_key	agent_id	Yes	Enforces uniqueness of agent_id
app_user	app_user_user_type_status_idx	user_type, status	No	Supports filtering and sorting on user_type, status
app_user	app_user_username_key	username	Yes	Enforces uniqueness of username
password_history	password_history_pkey	id	Yes	Primary key
password_history	password_history_user_id_created_at_idx	user_id, created_at	No	Finds child rows of a parent; supports the foreign key
role	role_pkey	id	Yes	Primary key
role	role_code_key	code	Yes	Enforces uniqueness of code
role_permission	role_permission_pkey	role_id, permission	Yes	Primary key
user_role	user_role_pkey	user_id, role_id	Yes	Primary key
user_session	user_session_pkey	sid	Yes	Primary key
user_session	user_session_expire_idx	expire	No	Supports filtering and sorting on expire
agency	agency_pkey	id	Yes	Primary key
agency	agency_code_key	code	Yes	Enforces uniqueness of code
agent	agent_pkey	id	Yes	Primary key
agent	agent_agency_id_status_idx	agency_id, status	No	Finds child rows of a parent; supports the foreign key
agent	agent_agent_code_key	agent_code	Yes	Enforces uniqueness of agent_code
agent	agent_id_type_id_number_hash_key	id_type, id_number_hash	Yes	Enforces uniqueness of id_type + id_number_hash
agent	agent_parent_agent_id_idx	parent_agent_id	No	Finds child rows of a parent; supports the foreign key
participant	participant_pkey	id	Yes	Primary key
participant	participant_created_by_agency_id_idx	created_by_agency_id	No	Supports filtering and sorting on created_by_agency_id
participant	participant_full_name_idx	full_name	No	Supports filtering and sorting on full_name
participant	participant_id_type_id_number_hash_key	id_type, id_number_hash	Yes	Enforces uniqueness of id_type + id_number_hash
participant	participant_participant_no_key	participant_no	Yes	Enforces uniqueness of participant_no
aml_screening	aml_screening_pkey	id	Yes	Primary key
aml_screening	aml_screening_status_idx	status	No	Supports filtering and sorting on status
aml_screening	aml_screening_subject_type_subject_id_idx	subject_type, subject_id	No	Supports filtering and sorting on subject_type, subject_id
aml_watchlist_entry	aml_watchlist_entry_pkey	id	Yes	Primary key
aml_watchlist_entry	aml_watchlist_entry_normalised_idx	normalised	No	Supports filtering and sorting on normalised

Table	Index	Columns	Unique	Purpose
product	product_pkey	id	Yes	Primary key
product	product_code_key	code	Yes	Enforces uniqueness of code
policy	policy_pkey	id	Yes	Primary key
policy	policy_agency_id_status_idx	agency_id, status	No	Finds child rows of a parent; supports the foreign key
policy	policy_agent_id_status_idx	agent_id, status	No	Finds child rows of a parent; supports the foreign key
policy	policy_issued_at_idx	issued_at	No	Supports filtering and sorting on issued_at
policy	policy_participant_id_idx	participant_id	No	Finds child rows of a parent; supports the foreign key
policy	policy_policy_no_key	policy_no	Yes	Enforces uniqueness of policy_no
policy	policy_quotation_no_key	quotation_no	Yes	Enforces uniqueness of quotation_no
policy	policy_status_payment_status_payment_due_date_idx	status, payment_status, payment_due_date	No	Supports filtering and sorting on status, payment_status, payment_due_date
nominee	nominee_pkey	id	Yes	Primary key
nominee	nominee_policy_id_idx	policy_id	No	Finds child rows of a parent; supports the foreign key
policy_event	policy_event_pkey	id	Yes	Primary key
policy_event	policy_event_policy_id_created_at_idx	policy_id, created_at	No	Finds child rows of a parent; supports the foreign key
signature_request	signature_request_pkey	id	Yes	Primary key
signature_request	signature_request_policy_id_idx	policy_id	No	Finds child rows of a parent; supports the foreign key
signature_request	signature_request_token_hash_key	token_hash	Yes	Enforces uniqueness of token_hash
payment	payment_pkey	id	Yes	Primary key
payment	payment_agency_id_status_idx	agency_id, status	No	Finds child rows of a parent; supports the foreign key
payment	payment_payment_no_key	payment_no	Yes	Enforces uniqueness of payment_no
payment	payment_status_created_at_idx	status, created_at	No	Supports filtering and sorting on status, created_at
payment_allocation	payment_allocation_pkey	id	Yes	Primary key
payment_allocation	payment_allocation_payment_id_policy_id_key	payment_id, policy_id	Yes	Enforces uniqueness of payment_id + policy_id
payment_allocation	payment_allocation_policy_id_idx	policy_id	No	Finds child rows of a parent; supports the foreign key
receipt	receipt_pkey	id	Yes	Primary key
receipt	receipt_issued_at_idx	issued_at	No	Supports filtering and sorting on issued_at
receipt	receipt_policy_id_idx	policy_id	No	Finds child rows of a parent; supports the foreign key
receipt	receipt_receipt_no_key	receipt_no	Yes	Enforces uniqueness of receipt_no
commission	commission_pkey	id	Yes	Primary key
commission	commission_agent_id_period_idx	agent_id, period	No	Finds child rows of a parent; supports the foreign key
commission	commission_policy_id_agent_id_key	policy_id, agent_id	Yes	Enforces uniqueness of policy_id + agent_id
claim	claim_pkey	id	Yes	Primary key
claim	claim_agency_id_status_idx	agency_id, status	No	Supports filtering and sorting on agency_id, status
claim	claim_claim_no_key	claim_no	Yes	Enforces uniqueness of claim_no

Table	Index	Columns	Unique	Purpose
claim	claim_policy_id_idx	policy_id	No	Finds child rows of a parent; supports the foreign key
document	document_pkey	id	Yes	Primary key
document	document_owner_type_owner_id_idx	owner_type, owner_id	No	Supports filtering and sorting on owner_type, owner_id
document	document_status_idx	status	No	Supports filtering and sorting on status
document	document_storage_key_key	storage_key	Yes	Enforces uniqueness of storage_key
workflow_definition	workflow_definition_pkey	id	Yes	Primary key
workflow_definition	workflow_definition_type_key	type	Yes	Enforces uniqueness of type
workflow_step	workflow_step_pkey	id	Yes	Primary key
workflow_step	workflow_step_definition_id_level_key	definition_id, level	Yes	Enforces uniqueness of definition_id + level
approval_request	approval_request_pkey	id	Yes	Primary key
approval_request	approval_request_entity_type_entity_id_idx	entity_type, entity_id	No	Supports filtering and sorting on entity_type, entity_id
approval_request	approval_request_maker_id_idx	maker_id	No	Supports filtering and sorting on maker_id
approval_request	approval_request_request_no_key	request_no	Yes	Enforces uniqueness of request_no
approval_request	approval_request_status_type_idx	status, type	No	Supports filtering and sorting on status, type
approval_action	approval_action_pkey	id	Yes	Primary key
approval_action	approval_action_request_id_idx	request_id	No	Finds child rows of a parent; supports the foreign key
issue	issue_pkey	id	Yes	Primary key
issue	issue_assigned_to_id_idx	assigned_to_id	No	Supports filtering and sorting on assigned_to_id
issue	issue_issue_no_key	issue_no	Yes	Enforces uniqueness of issue_no
issue	issue_reported_by_id_idx	reported_by_id	No	Supports filtering and sorting on reported_by_id
issue	issue_status_priority_idx	status, priority	No	Supports filtering and sorting on status, priority
issue_comment	issue_comment_pkey	id	Yes	Primary key
issue_comment	issue_comment_issue_id_idx	issue_id	No	Finds child rows of a parent; supports the foreign key
notification	notification_pkey	id	Yes	Primary key
notification	notification_status_channel_idx	status, channel	No	Supports filtering and sorting on status, channel
notification	notification_user_id_channel_read_at_idx	user_id, channel, read_at	No	Finds child rows of a parent; supports the foreign key
audit_log	audit_log_pkey	id	Yes	Primary key
audit_log	audit_log_action_idx	action	No	Supports filtering and sorting on action
audit_log	audit_log_actor_id_occurred_at_idx	actor_id, occurred_at	No	Supports filtering and sorting on actor_id, occurred_at
audit_log	audit_log_entity_type_entity_id_idx	entity_type, entity_id	No	Supports filtering and sorting on entity_type, entity_id
audit_log	audit_log_occurred_at_idx	occurred_at	No	Supports filtering and sorting on occurred_at
config_parameter	config_parameter_pkey	key	Yes	Primary key
code_item	code_item_pkey	id	Yes	Primary key
code_item	code_item_category_code_key	category, code	Yes	Enforces uniqueness of category + code
outbox_message	outbox_message_pkey	id	Yes	Primary key

Table	Index	Columns	Unique	Purpose
outbox_message	outbox_message_status_next_attempt_at_idx	status, next_attempt_at	No	Supports filtering and sorting on status, next_attempt_at
outbox_message	outbox_message_system_created_at_idx	system, created_at	No	Supports filtering and sorting on system, created_at
integration_log	integration_log_pkey	id	Yes	Primary key
integration_log	integration_log_success_created_at_idx	success, created_at	No	Supports filtering and sorting on success, created_at
integration_log	integration_log_system_created_at_idx	system, created_at	No	Supports filtering and sorting on system, created_at
reconciliation_run	reconciliation_run_pkey	id	Yes	Primary key
reconciliation_run	reconciliation_run_business_date_system_idx	business_date, system	No	Supports filtering and sorting on business_date, system
eod_run	eod_run_pkey	id	Yes	Primary key
eod_run	eod_run_business_date_key	business_date	Yes	Enforces uniqueness of business_date
report_schedule	report_schedule_pkey	id	Yes	Primary key
report_schedule	report_schedule_active_next_run_at_idx	active, next_run_at	No	Supports filtering and sorting on active, next_run_at

Table 4: Indexes

5. Enumerations

Enumerations are PostgreSQL ENUM types. Adding a value requires a migration; values are never removed while rows use them.

UserType: kind of user account.

Value	Meaning
AGENT	Agency agent or sub-agent using the portal.
BANCA	Bank officer using the portal.
STAFF	IIFT staff using the back-office.

Used by: app_user.user_type

UserStatus: state of a user account.

Value	Meaning
ACTIVE	Can sign in.
LOCKED	Temporarily locked after repeated failed sign-ins; unlocks at locked_until or by an administrator.
DISABLED	Cannot sign in until re-activated.

Used by: app_user.status

AuthSource: where the password is verified.

Value	Meaning
LOCAL	Argon2id hash held in app_user.
DIRECTORY	Active Directory / LDAP bind.

Used by: app_user.auth_source

Audience: application a role or user belongs to.

Value	Meaning
PORTAL	Agent/Banca Portal.
BACKOFFICE	Back-office.

Used by: role.audience

Channel: distribution channel of an agency.

Value	Meaning
AGENCY	Agency channel.
BANCA	Bancatakaful channel (bank partner).

Used by: agency.channel

AgencyStatus: state of an agency or bank.

Value	Meaning
ACTIVE	Trading; users can sign in.
INACTIVE	Not trading; users cannot sign in.
SUSPENDED	Suspended by IIFT; users cannot sign in.

Used by: agency.status

AgentType: position in the distribution hierarchy.

Value	Meaning
MAIN_AGENT	Main agent; may have sub-agents.
SUB_AGENT	Sub-agent reporting to a main agent.
BANKER	Bank officer.

Used by: agent.agent_type

AgentStatus: registration status of an agent.

Value	Meaning
PENDING	Registration awaiting approval.
ACTIVE	Approved and able to transact.
INACTIVE	Deactivated; cannot sign in.
SUSPENDED	Suspended; cannot sign in.
TERMINATED	Contract ended.
REJECTED	Registration rejected.

Used by: agent.status

IdType: identification document type.

Value	Meaning
NRIC	Brunei identity card (IC).
PASSPORT	Passport.
BUSINESS_REG	Business registration number of a company.

Used by: agent.id_type, participant.id_type

ParticipantType: kind of participant.

Value	Meaning
INDIVIDUAL	Natural person.
CORPORATE	Company or organisation.

Used by: participant.type

AmlStatus: aML/KYC outcome held on agents and participants.

Value	Meaning
NOT_SCREENED	Not yet screened.
CLEAR	No match, or match cleared by Compliance.
FLAGGED	Potential match awaiting Compliance review; blocks submission.
REJECTED	Match confirmed by Compliance; business is refused.

Used by: agent.aml_status, participant.aml_status

AmlCaseStatus: state of a screening case.

Value	Meaning
AUTO_CLEARED	Score below the threshold; no review needed.
PENDING_REVIEW	Score at or above the threshold; awaiting Compliance.
CLEARED	Compliance decided the match is false.
CONFIRMED_MATCH	Compliance confirmed the match.

Used by: aml_screening.status

PolicyStatus: lifecycle status of a quotation or policy.

Value	Meaning
DRAFT	Quotation being prepared; editable.
PENDING_APPROVAL	Referred to IIFT for approval or quality check.
PENDING_PAYMENT	Accepted; pay-first product awaiting verified payment before issuance.
ACTIVE	Issued and in force.
REJECTED	Rejected by a checker; can be reopened as a draft.
EXPIRED	Cover period ended.
CANCELLED	Cancelled after an approved cancellation request.

Used by: policy.status, policy_event.from_status, policy_event.to_status

PolicyPaymentStatus: payment state of a policy.

Value	Meaning
UNPAID	No verified payment and none pending.
PENDING_VERIFICATION	Payment submitted, awaiting Finance.
PAID	Fully paid.

Used by: policy.payment_status

NomineeRole: role of a person named on a policy.

Value	Meaning
NOMINEE	Nominee.
BENEFICIARY	Beneficiary.
EXECUTOR	Executor.

Used by: nominee.role

PaymentMethod: how a payment was made.

Value	Meaning
BANK_TRANSFER	Bank transfer.
CHEQUE	Cheque.
CASH_DEPOSIT	Cash deposited at a bank.
ONLINE	Online payment.

Used by: payment.method

PaymentStatus: verification state of a payment.

Value	Meaning
PENDING_VERIFICATION	Awaiting Finance verification.
VERIFIED	Verified; receipts issued.
REJECTED	Rejected with a reason.

Used by: payment.status

ClaimStatus: state of a claim notification.

Value	Meaning
SUBMITTED	Notified by the agent.
UNDER_REVIEW	Being reviewed by IIFT.
ACKNOWLEDGED	Accepted and registered in the core system.
REJECTED	Not accepted.
CLOSED	Closed.

Used by: claim.status

DocumentOwnerType: type of record a document is attached to.

Value	Meaning
AGENT	Agent profile.
AGENCY	Agency or bank.
PARTICIPANT	Participant.
POLICY	Quotation or policy.
PAYMENT	Payment.
CLAIM	Claim notification.
ISSUE	Issue.
REPORT	EOD run or report schedule.

Used by: document.owner_type

DocumentStatus: verification state of a document.

Value	Meaning
UPLOADED	Uploaded, not yet checked.
VERIFIED	Accepted by a checker.
REJECTED	Rejected by a checker with remarks.

Used by: document.status

ApprovalType: transaction types that run through maker-checker.

Value	Meaning
AGENT_REGISTRATION	New agent or bank officer.
AGENT_PROFILE_UPDATE	Change to agent details or reporting line.
AGENT_STATUS_CHANGE	Activation, suspension, deactivation or termination.
PARTICIPANT_UPDATE	Change to participant details.
POLICY_REFERRAL	Referred application or quality check before issuance.
POLICY_ENDORSEMENT	Policy endorsement.
POLICY_CANCELLATION	Policy cancellation with refund.
PAYMENT_VERIFICATION	Verification of a payment and receipt issuance.

Used by: workflow_definition.type, approval_request.type

ApprovalStatus: state of an approval request.

Value	Meaning
PENDING	Awaiting a decision at the current level.
APPROVED	All levels approved; the change has been applied.
REJECTED	Rejected at a level, with remarks.
WITHDRAWN	Withdrawn by the maker.

Used by: approval_request.status

ApprovalActionType: action recorded on an approval request.

Value	Meaning
SUBMIT	Request raised.
APPROVE	Level approved.

Value	Meaning
REJECT	Request rejected.
WITHDRAW	Request withdrawn.

Used by: approval_action.action

IssuePriority: issue priority; selects SLA hours.

Value	Meaning
CRITICAL	Service down or critical function unavailable.
HIGH	Major function impaired.
MEDIUM	Function impaired with a workaround.
LOW	Minor or cosmetic.

Used by: issue.priority

IssueStatus: state of an issue.

Value	Meaning
OPEN	Reported, not assigned.
ASSIGNED	Assigned to a person or team.
IN_PROGRESS	Being worked on.
RESOLVED	Resolution provided.
CLOSED	Confirmed and closed.

Used by: issue.status

NotificationChannel: delivery channel.

Value	Meaning
IN_APP	Bell icon in the portal or back-office.
EMAIL	E-mail through the SMTP relay.
SMS	SMS through the SMS gateway.

Used by: notification.channel

DeliveryStatus: delivery state of a notification.

Value	Meaning
PENDING	Waiting for delivery or retry.
SENT	Accepted by the gateway.
FAILED	Attempts exhausted.

Used by: notification.status

IntegrationSystem: external system of an integration message or log entry.

Value	Meaning
CORE	IITH core takaful system.
FINANCE	Financial system (FIN).
AML	External AML screening service.
EMAIL	SMTP relay.
SMS	SMS gateway.
DIRECTORY	Active Directory / LDAP.

Used by: outbox_message.system, integration_log.system, reconciliation_run.system

OutboxStatus: delivery state of an outbox message.

Value	Meaning
PENDING	Waiting for first delivery.
SENT	Delivered.
FAILED	Failed; retry scheduled at next_attempt_at.
DEAD	Dead-letter: attempts exhausted or error not retryable; support alerted.

Used by: outbox_message.status

IntegrationDirection: direction of an integration call.

Value	Meaning
OUTBOUND	Portal calls the external system.
INBOUND	External system calls the portal.

Used by: integration_log.direction

JobStatus: state of an end-of-day run.

Value	Meaning
RUNNING	In progress.
COMPLETED	Finished successfully.
FAILED	Finished with an error; can be re-run.

Used by: eod_run.status

ReconciliationStatus: outcome of a reconciliation run.

Value	Meaning
MATCHED	Every receipt was found in the financial system.
MISMATCH	One or more receipts were not found.

Used by: reconciliation_run.status

ParameterType: data type of a configuration parameter.

Value	Meaning
STRING	Text.
INTEGER	Whole number.
DECIMAL	Decimal number.
BOOLEAN	true or false.

Used by: config_parameter.value_type

ReportFrequency: how often a scheduled report runs.

Value	Meaning
DAILY	Every day at 07:00 Brunei time.
WEEKLY	Every Monday at 07:00.
MONTHLY	On the 1st of each month at 07:00.

Used by: report_schedule.frequency

ExportFormat: report output format.

Value	Meaning
XLSX	Excel workbook.
CSV	Comma-separated values.
PDF	PDF document.

Used by: report_schedule.format

CommissionStatus: payment state of a commission accrual.

Value	Meaning
ACCRUED	Accrued at issuance, not yet paid.
PAID	Payment confirmed by Finance.

Used by: commission.status

6. Other Database Objects

6.1 Sequences

Business reference numbers come from PostgreSQL sequences, which are safe under concurrency across API replicas. A rolled-back transaction can leave a gap; numbers are unique but not guaranteed contiguous.

Sequence	Used for	Column	Format example
seq_agent_code	Agent codes	agent.agent_code	AG-000045 (agency) or BK-000012 (bank officer)
seq_participant_no	Participant numbers	participant.participant_no	PT/26/000123
seq_quotation_no	Quotation numbers	policy.quotation_no	QT/26/000123
seq_policy_no	Policy numbers	policy.policy_no	PRO/26/000045 (product code prefix)
seq_payment_no	Payment references	payment.payment_no	PY/26/000123
seq_receipt_no	Receipt numbers	receipt.receipt_no	RC/26/000123
seq_claim_no	Claim notification numbers	claim.claim_no	CL/26/000123
seq_issue_no	Issue numbers	issue.issue_no	IS/26/000123
seq_request_no	Approval request numbers	approval_request.request_no	RQ/26/000123

Table 5: Sequences

6.2 Audit trail trigger

Function **audit_log_block_mutation** (returns trigger) raises an exception; trigger **audit_log_no_update** calls it BEFORE UPDATE OR DELETE on audit_log, for each row. Any attempt to change or delete audit rows fails, whichever client issues it. Removing audit rows at the end of the retention period is done by a controlled archive procedure run by the database owner with the trigger disabled inside one audited maintenance transaction (delivered during implementation).

```
CREATE FUNCTION "audit_log_block_mutation"() RETURNS trigger AS $$
BEGIN
    RAISE EXCEPTION 'audit_log is append-only';
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER "audit_log_no_update"
BEFORE UPDATE OR DELETE ON "audit_log"
FOR EACH ROW EXECUTE FUNCTION "audit_log_block_mutation"();
```

6.3 Check constraints

Constraint	Table	Expression	Rule
policy_amounts_non_negative	policy	"contribution" >= 0 AND "sum_covered" >= 0 AND "outstanding_amount" >= 0	Contribution, sum covered and outstanding amount can never be negative.
payment_amount_positive	payment	"total_amount" > 0	A payment must be for a positive amount.
payment_allocation_amount_positive	payment_allocation	"amount" > 0	Each allocation of a payment to a policy must be positive.
nominee_share_range	nominee	"share_percent" > 0 AND "share_percent" <= 100	A nominee's share is greater than 0 and at most 100 per cent.

Table 6: Check constraints

6.4 Session table

user_session has the structure expected by connect-pg-simple (sid, sess, expire). It is created by the migration, not by the library (createTableIfMissing is false), so the application database role needs no DDL rights. Expired sessions are pruned every 15 minutes.

7. Protected Data

Identification numbers are never stored in clear. The application encrypts them with AES-256-GCM using FIELD_ENCRYPTION_KEY before they reach the database, with a random 96-bit IV per value; the stored form is v1:<base64 of IV, authentication tag and ciphertext>, and the version prefix allows key rotation. To find a record by identification number and to enforce uniqueness, a blind index is stored alongside: an HMAC-SHA256 of the normalised number (spaces, hyphens and slashes removed, upper case) keyed with FIELD_HASH_KEY. The unique constraints (id_type, id_number_hash) on agent and participant therefore prevent duplicates without decrypting. Screens show only the last four characters.

Table	Column	Protection
agent	id_number_enc	Encrypted (AES-256-GCM, FIELD_ENCRYPTION_KEY)
participant	id_number_enc	Encrypted (AES-256-GCM, FIELD_ENCRYPTION_KEY)
nominee	id_number_enc	Encrypted (AES-256-GCM, FIELD_ENCRYPTION_KEY)
agent	id_number_hash	Blind index (HMAC-SHA256, FIELD_HASH_KEY)
participant	id_number_hash	Blind index (HMAC-SHA256, FIELD_HASH_KEY)
app_user	password_hash	One-way hash (Argon2id)
password_history	password_hash	One-way hash (Argon2id)
signature_request	token_hash	One-way hash (SHA-256 of one-time token)
user_session	sess	Holds the CSRF token; never exposed to the browser
notification	body	Removed after delivery when sensitive = true

Table 7: Protected columns

Document files are encrypted separately with DOCUMENT_ENCRYPTION_KEY (AES-256-GCM, one IV per file); the document table holds only metadata and a generated storage key. The audit service writes a fixed REDACTED marker in place of passwords, hashes, encrypted identifiers, tokens and storage keys.